

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

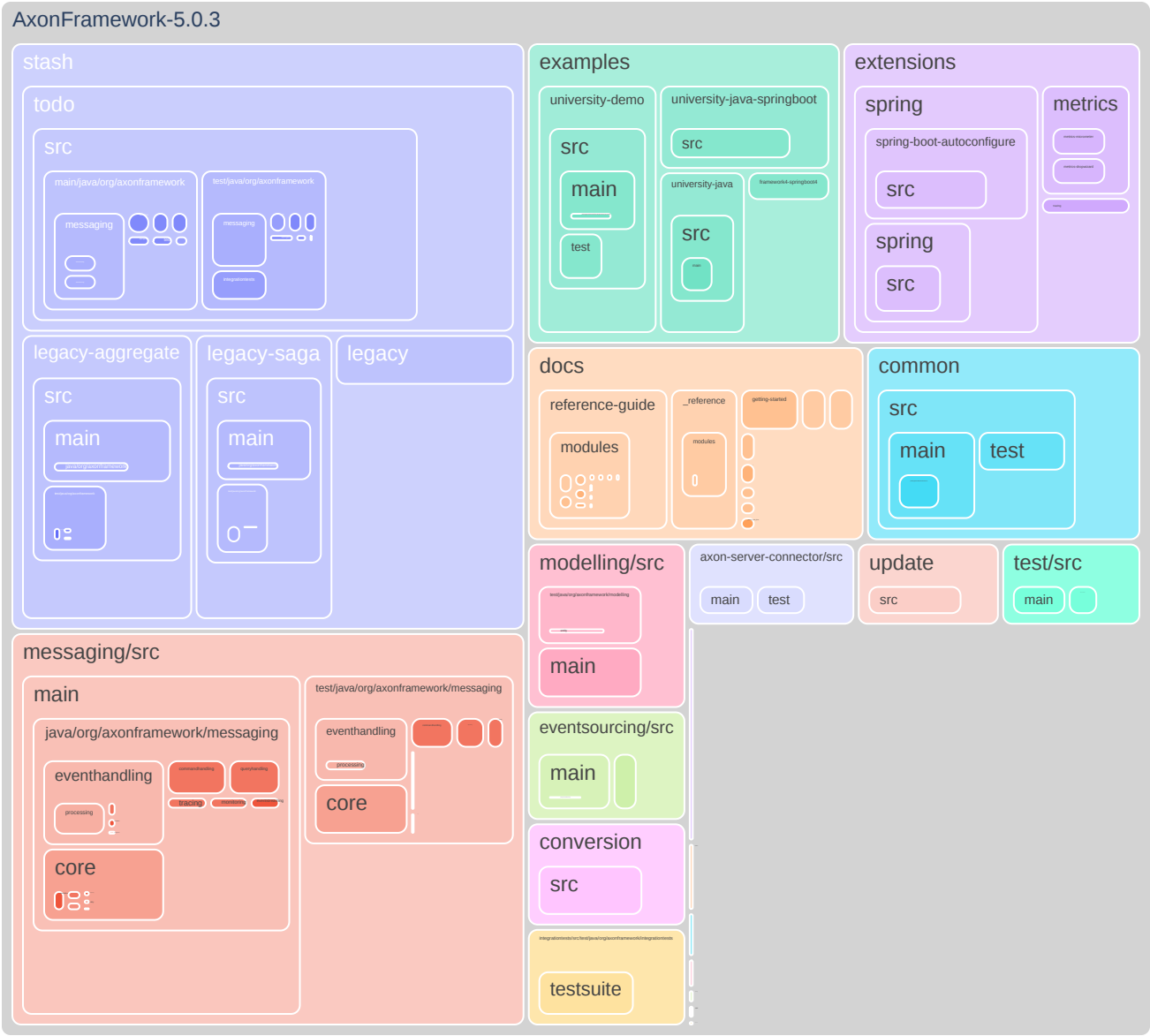
Data Preview

	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	790.000000	790.000000	790.000000	790.000000	790.000000	790.000000
mean	21.870886	11.225316	166.854430	57.267089	193.781013	152.581013
std	112.153100	12.024368	359.925155	6.964935	185.168868	159.942798
min	1.000000	2.000000	4.000000	46.000000	50.000000	46.000000
25%	2.000000	4.000000	23.000000	55.000000	102.000000	55.000000
50%	4.000000	7.000000	46.000000	57.000000	154.000000	106.000000
75%	11.000000	12.000000	155.000000	57.000000	166.000000	155.000000
max	2771.000000	117.000000	6350.000000	76.000000	2836.000000	683.000000

	directoryPath	directoryParentPath	directoryName
0	AxonFramework-5.0.3/.idea/copyright	AxonFramework-5.0.3/.idea	copyright
1	AxonFramework-5.0.3/.idea/inspectionProfiles	AxonFramework-5.0.3/.idea	inspectionProfile
2	AxonFramework-5.0.3/axon-framework-bom	AxonFramework-5.0.3	axon-framework-bom
3	AxonFramework-5.0.3/axon-server-connector/src/main/resources/META-INF/services	AxonFramework-5.0.3/axon-server-connector/src/main	resources/META-INF/service
4	AxonFramework-5.0.3/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/command	AxonFramework-5.0.3/axon-server-connector/src/test/java/org/axonframework/axonserver/connector	command
5	AxonFramework-5.0.3/axon-server-connector/src/test/resources	AxonFramework-5.0.3/axon-server-connector/src/test	resource
6	AxonFramework-5.0.3/build/coverage-report	AxonFramework-5.0.3/build	coverage-report
7	AxonFramework-5.0.3/build/parent	AxonFramework-5.0.3/build	parent
8	AxonFramework-5.0.3/common/src/main/java/org/axonframework/common/digest	AxonFramework-5.0.3/common/src/main/java/org/axonframework/common	digest
9	AxonFramework-5.0.3/common/src/main/java/org/axonframework/common/io	AxonFramework-5.0.3/common/src/main/java/org/axonframework/common	io
10	AxonFramework-5.0.3/common/src/main/java/org/axonframework/common/tx	AxonFramework-5.0.3/common/src/main/java/org/axonframework/common	tx
11	AxonFramework-5.0.3/common/src/main/resources/META-INF/services	AxonFramework-5.0.3/common/src/main	resources/META-INF/service
12	AxonFramework-5.0.3/common/src/test/java/org/axonframework/common/lifecycle	AxonFramework-5.0.3/common/src/test/java/org/axonframework/common	lifecycle
13	AxonFramework-5.0.3/common/src/test/resources/META-INF/services	AxonFramework-5.0.3/common/src/test	resources/META-INF/service
14	AxonFramework-5.0.3/conversion/src/main/resources/META-INF/services	AxonFramework-5.0.3/conversion/src/main	resources/META-INF/service
15	AxonFramework-5.0.3/docs/_playbook/localLinks	AxonFramework-5.0.3/docs/_playbook	localLink
16	AxonFramework-5.0.3/docs/_reference/modules/ROOT/attachments	AxonFramework-5.0.3/docs/_reference/modules/ROOT	attachment
17	AxonFramework-5.0.3/docs/_reference/modules/ROOT/partials	AxonFramework-5.0.3/docs/_reference/modules/ROOT	partial
18	AxonFramework-5.0.3/docs/_reference/modules/axon-server-connector/pages	AxonFramework-5.0.3/docs/_reference/modules/axon-server-connector	page
19	AxonFramework-5.0.3/docs/_reference/modules/axon-server-connector/partials	AxonFramework-5.0.3/docs/_reference/modules/axon-server-connector	partial
20	AxonFramework-5.0.3/docs/_reference/modules/configuration/pages	AxonFramework-5.0.3/docs/_reference/modules/configuration	page
21	AxonFramework-5.0.3/docs/_reference/modules/configuration/partials	AxonFramework-5.0.3/docs/_reference/modules/configuration	partial
22	AxonFramework-5.0.3/docs/_reference/modules/disruptor/pages	AxonFramework-5.0.3/docs/_reference/modules/disruptor	page
23	AxonFramework-5.0.3/docs/_reference/modules/disruptor/partials	AxonFramework-5.0.3/docs/_reference/modules/disruptor	partial
24	AxonFramework-5.0.3/docs/_reference/modules/eventsourcing/pages	AxonFramework-5.0.3/docs/_reference/modules/eventsourcing	page
25	AxonFramework-5.0.3/docs/_reference/modules/eventsourcing/partials	AxonFramework-5.0.3/docs/_reference/modules/eventsourcing	partial
26	AxonFramework-5.0.3/docs/_reference/modules/legacy/pages	AxonFramework-5.0.3/docs/_reference/modules/legacy	page
27	AxonFramework-5.0.3/docs/_reference/modules/legacy/partials	AxonFramework-5.0.3/docs/_reference/modules/legacy	partial
28	AxonFramework-5.0.3/docs/_reference/modules/metrics/pages	AxonFramework-5.0.3/docs/_reference/modules/metrics	page
29	AxonFramework-5.0.3/docs/_reference/modules/metrics/partials	AxonFramework-5.0.3/docs/_reference/modules/metrics	partial

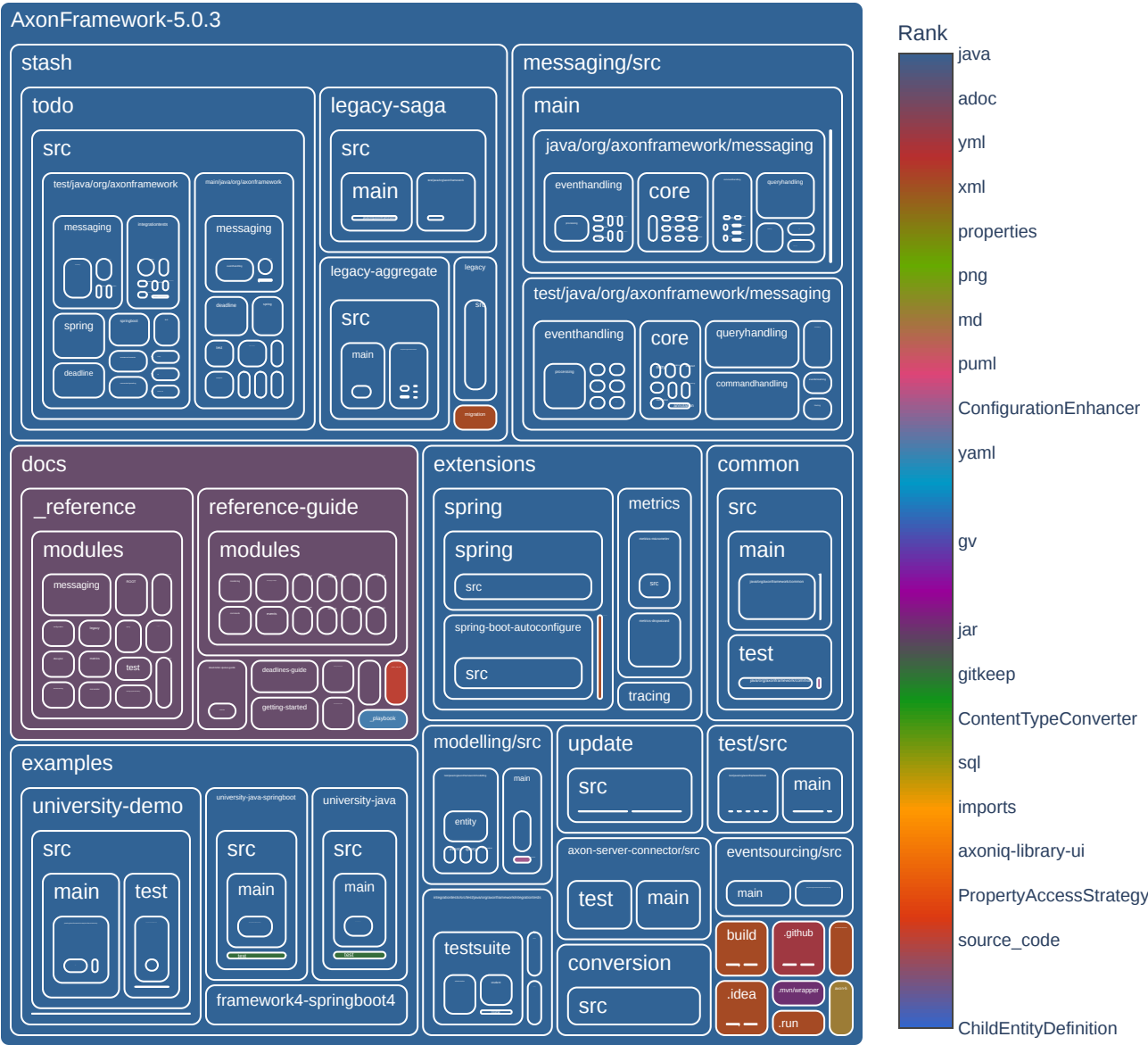
Number of files per directory

Directories and their file count



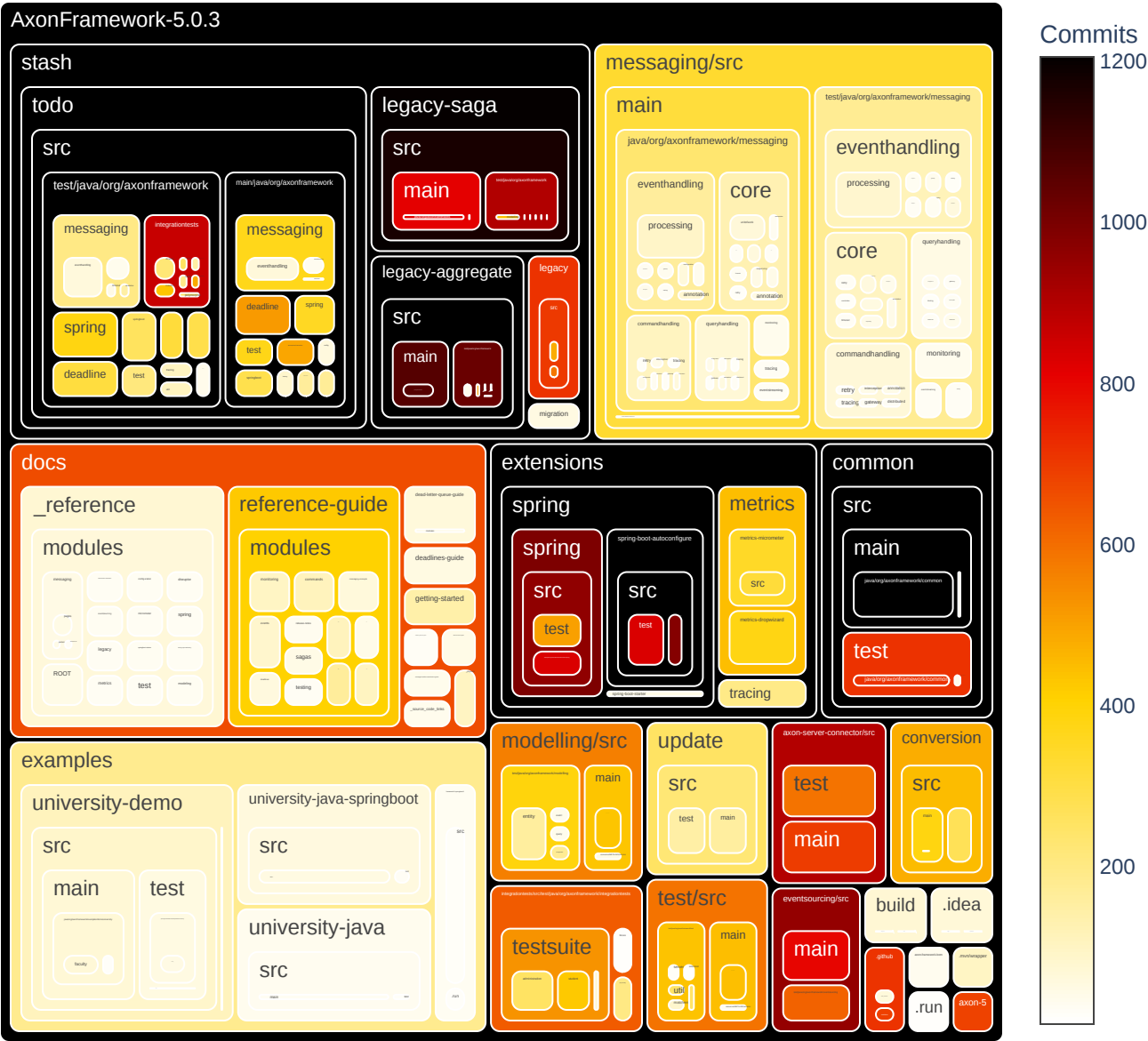
Most frequent file extension per directory

Most frequent file extension per directory



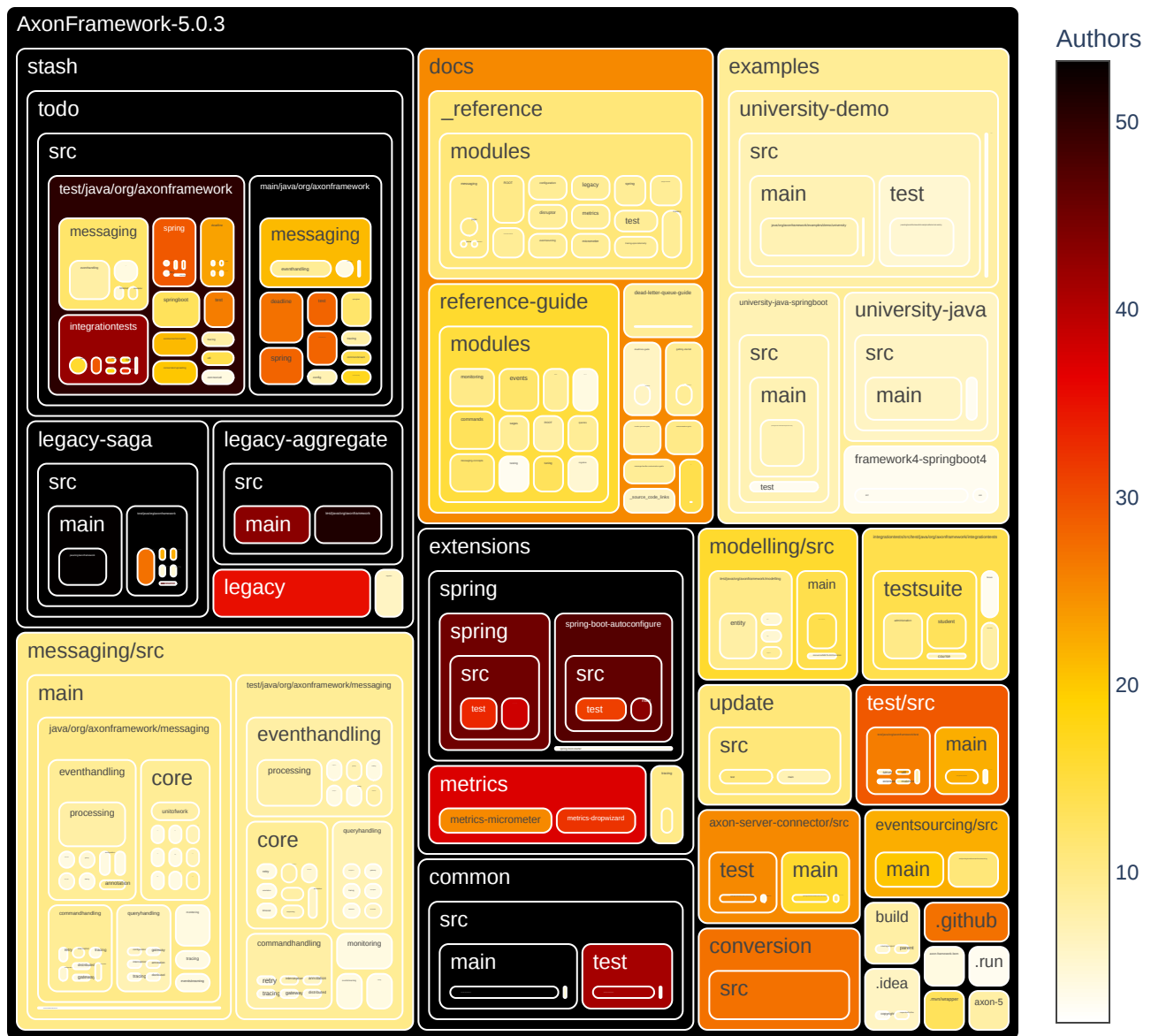
Number of commits per directory

Number of git commits



Number of distinct authors per directory

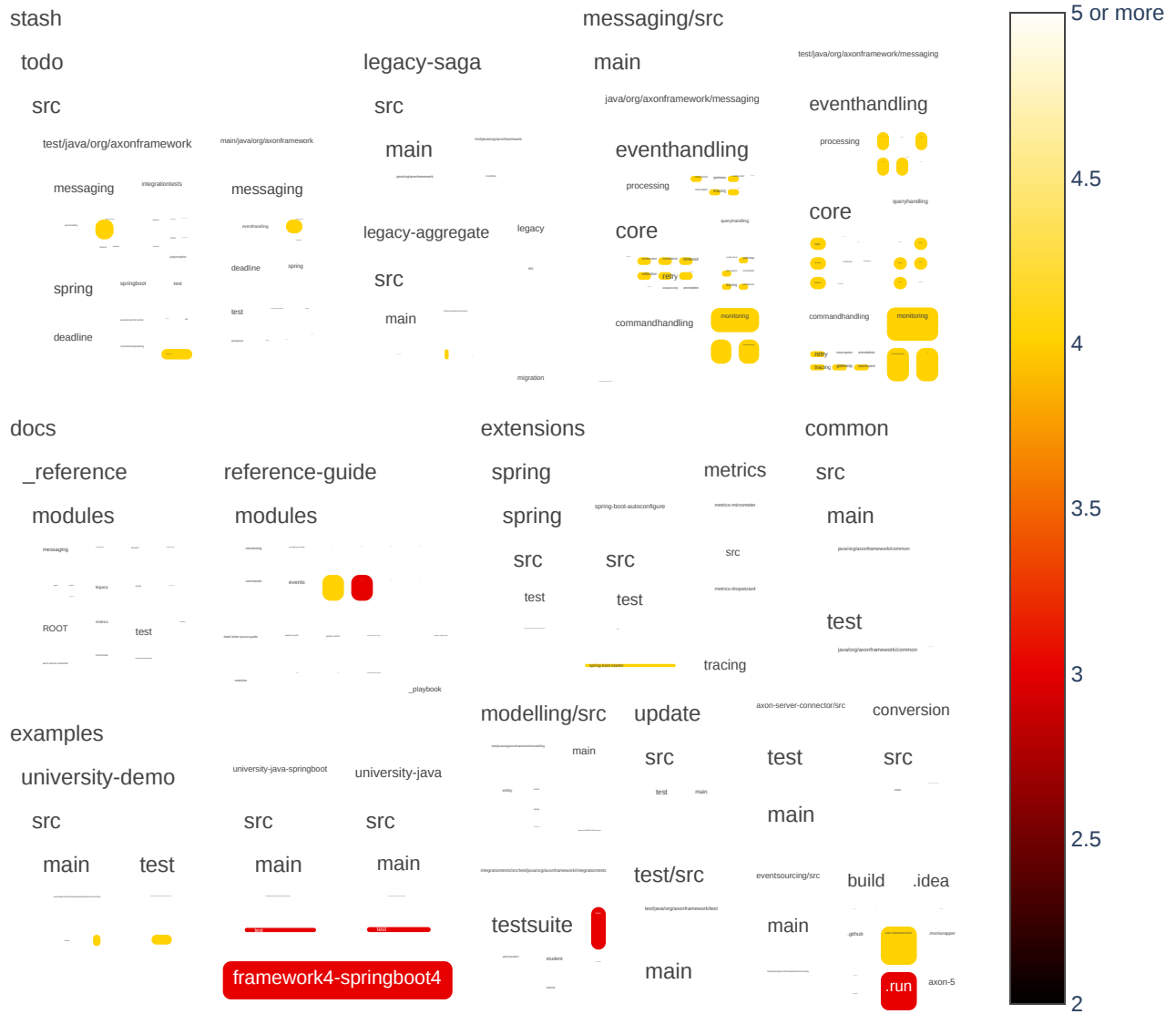
Number of distinct commit authors



Directories with very few different authors

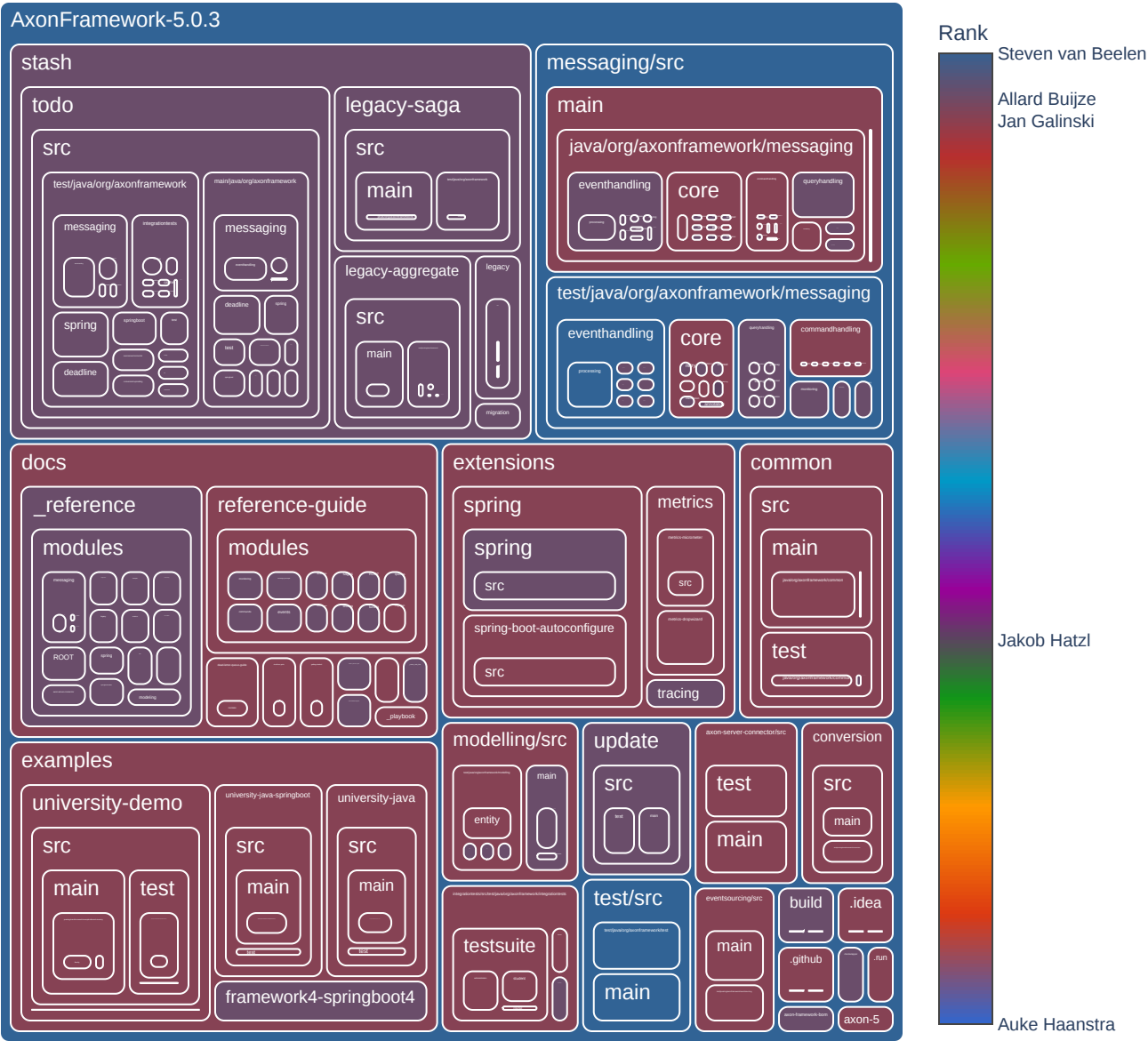
Number of distinct commit authors (red/black = only one or very few authors)

AxonFramework-5.0.3



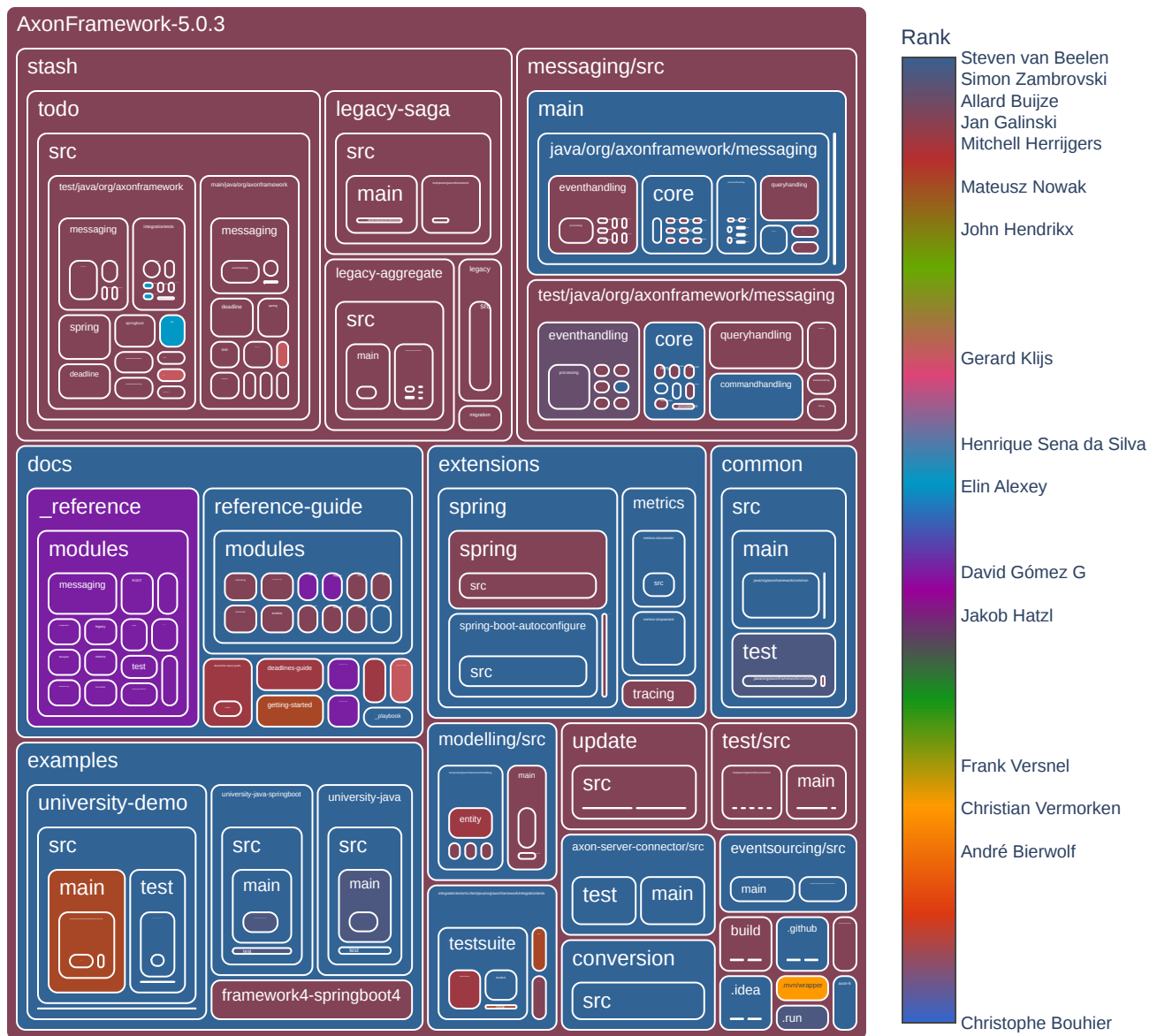
Main author per directory

Main authors with highest number of commits



Second author per directory

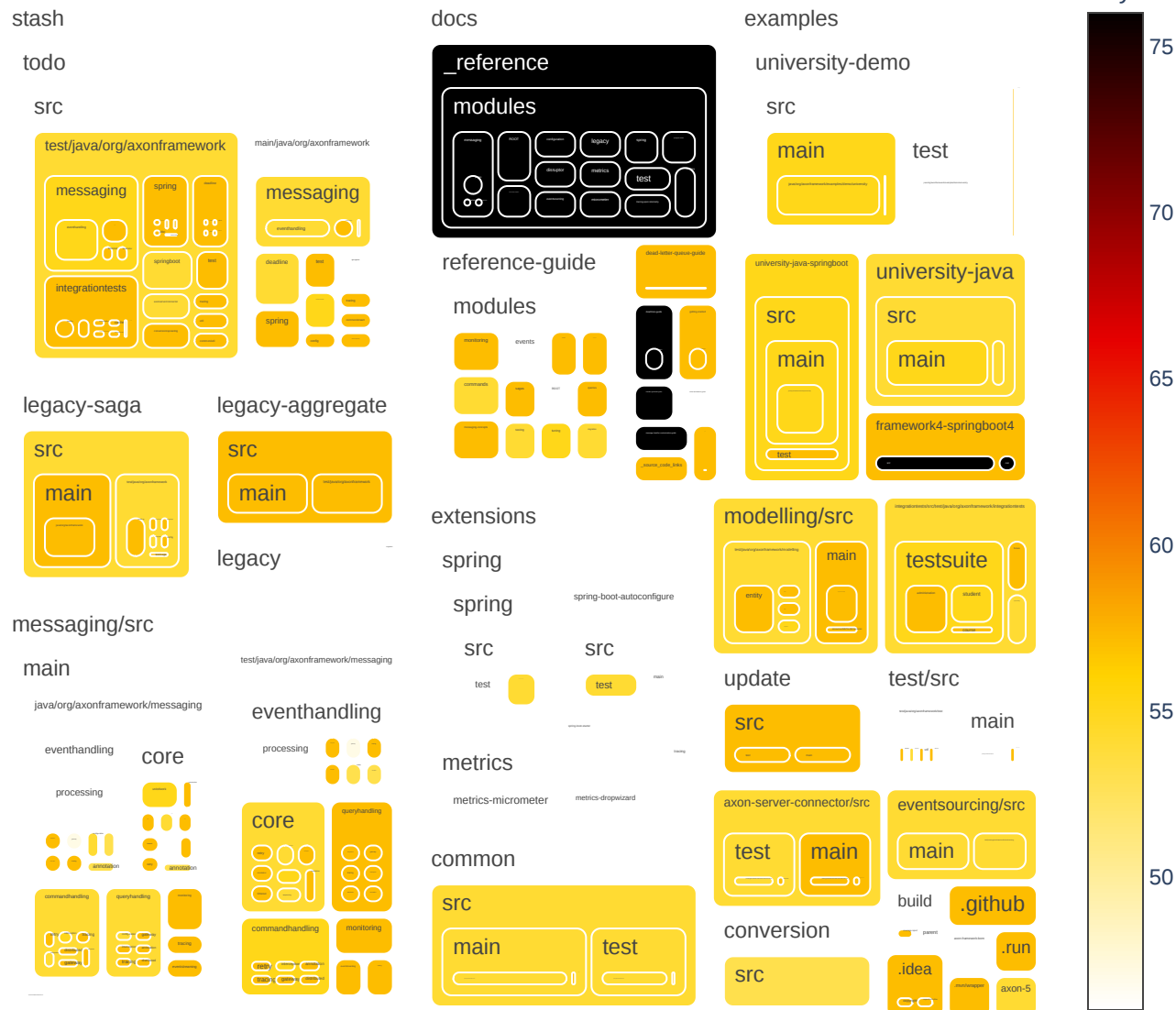
Second author with the second highest number of commits



Days since last commit per directory

Days since last commit

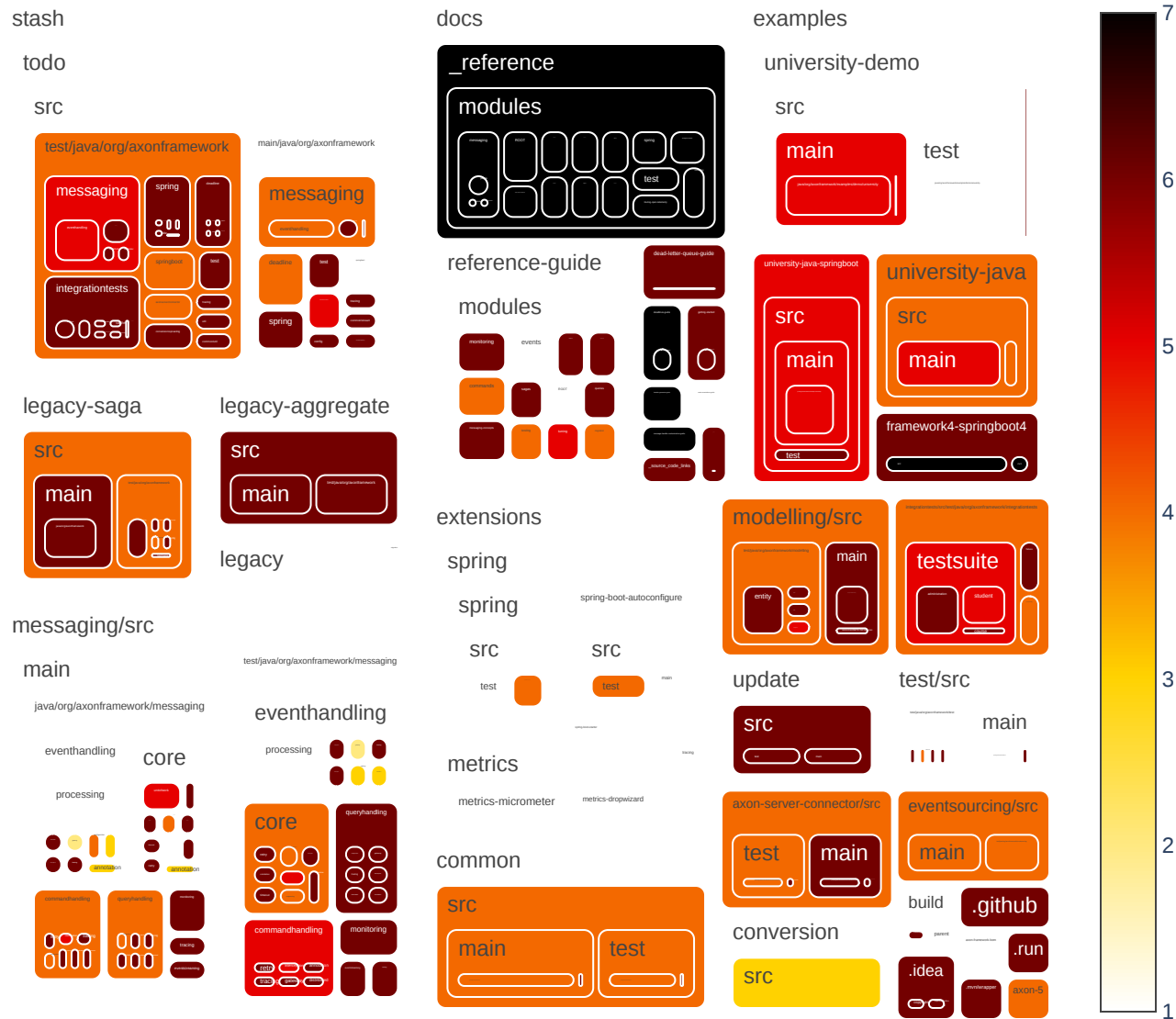
AxonFramework-5.0.3



Days since last commit per directory (ranked)

Rank of days since last commit

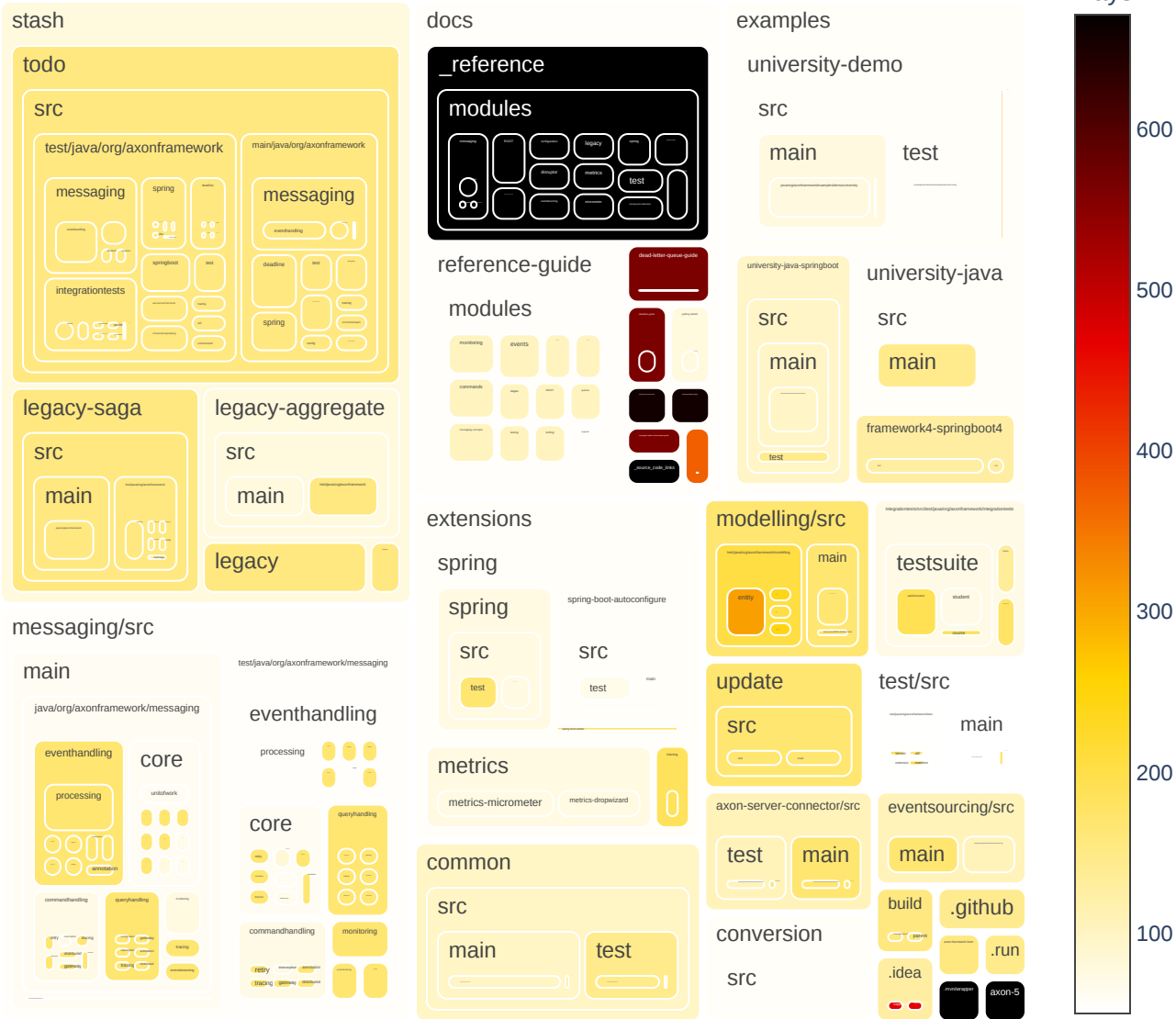
AxonFramework-5.0.3



Days since last file creation per directory

Days since last file creation

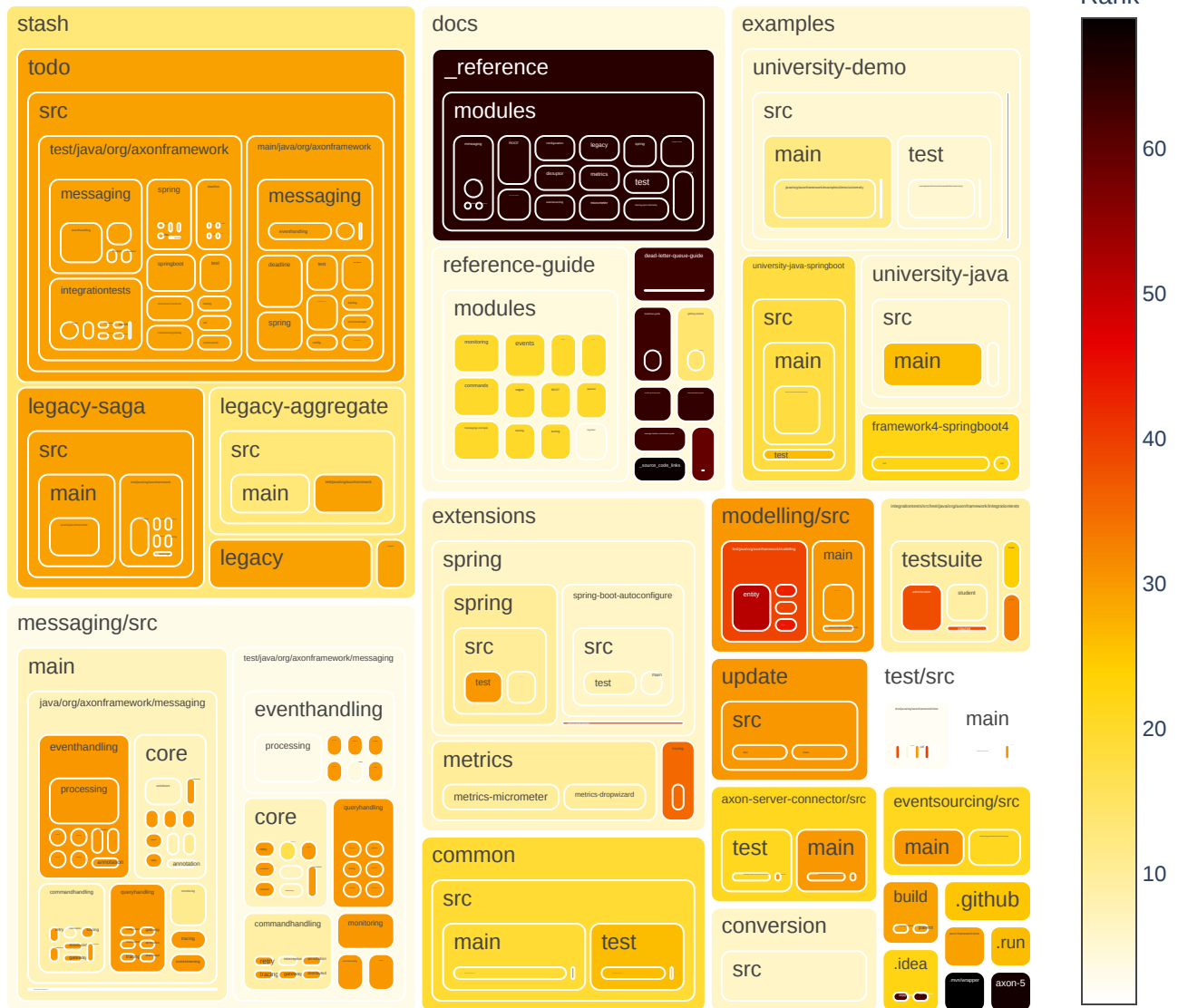
AxonFramework-5.0.3



Days since last file creation per directory (ranked)

Rank of days since last file creation

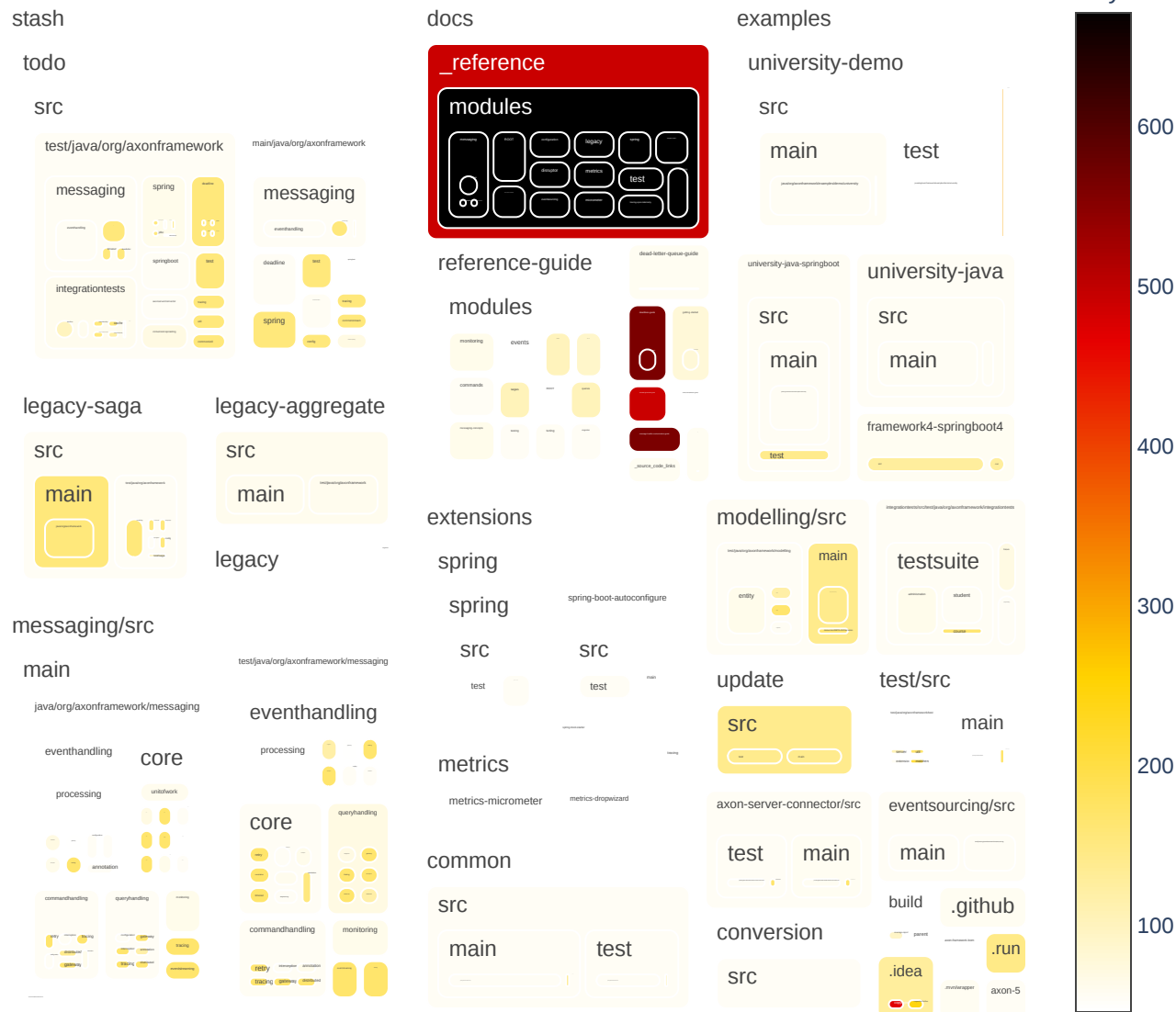
AxonFramework-5.0.3



Days since last file modification per directory

Days since last file modification

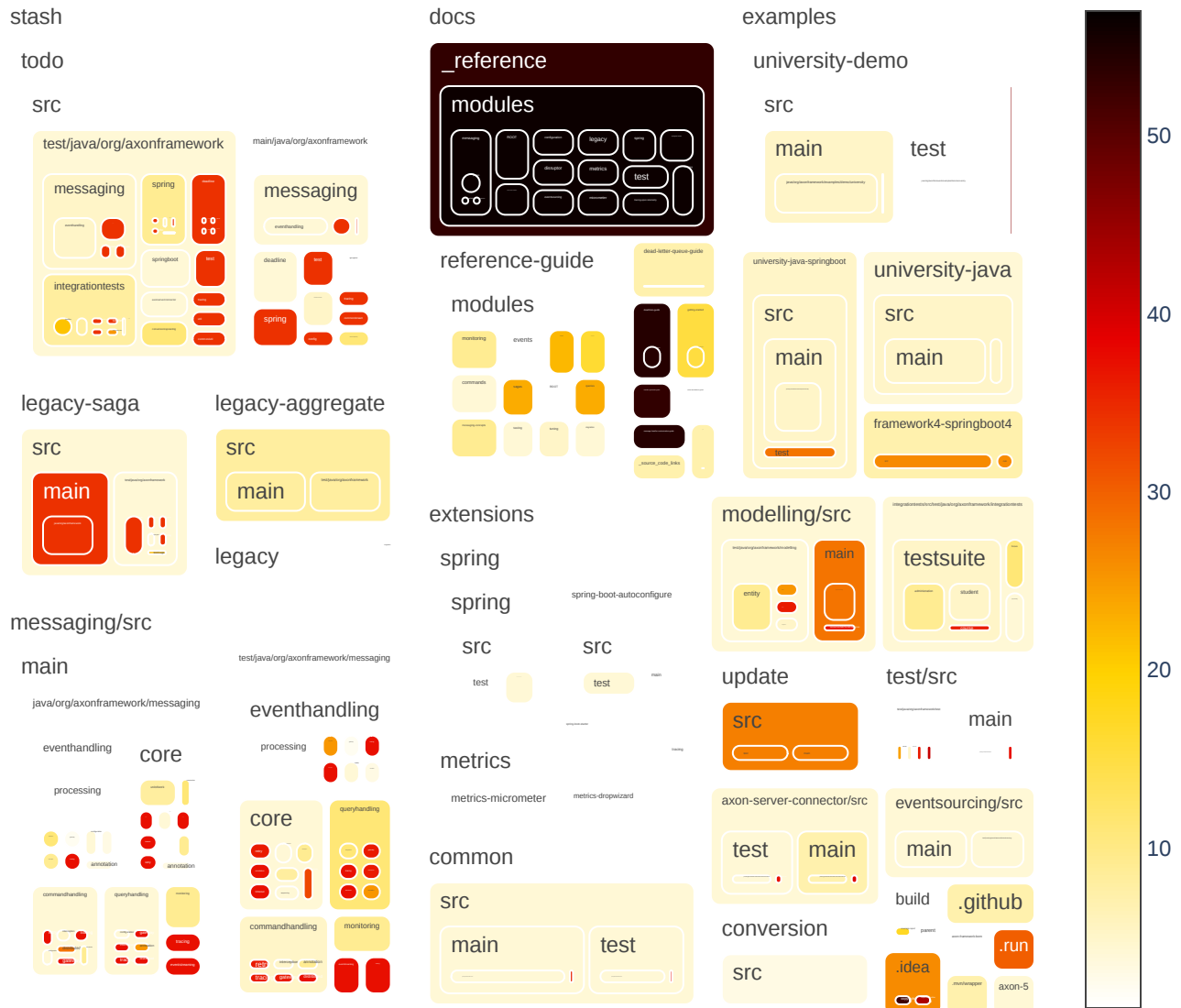
AxonFramework-5.0.3



Days since last file modification per directory (ranked)

Rank of days since last file modification

AxonFramework-5.0.3



Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

Preview data

Sum of commits that changed more than 30 files (each) = 1723

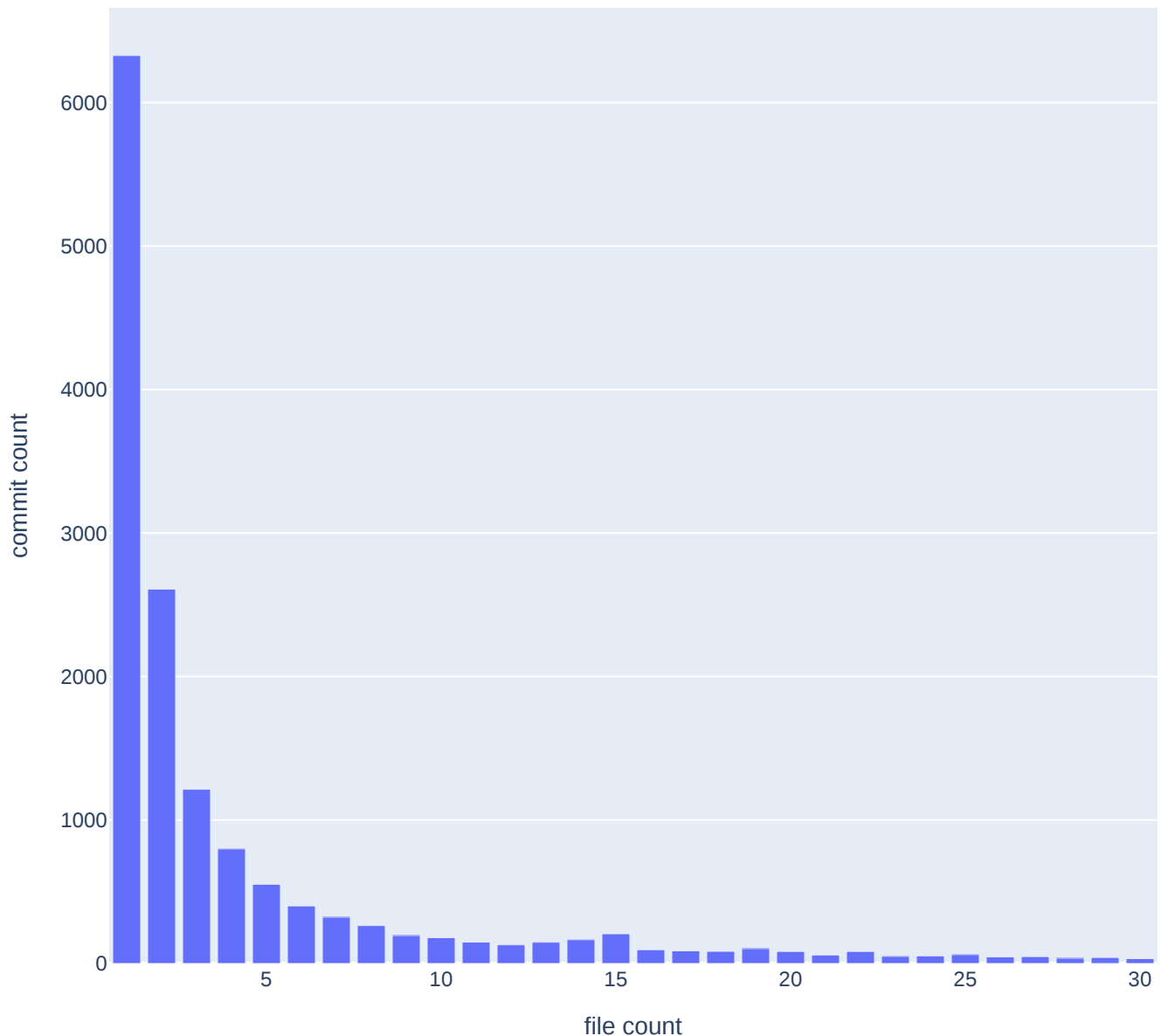
Max changed files with one commit = 4949

	filesPerCommit	commitCount
count	478.000000	478.000000
mean	585.619247	34.223849
std	854.225728	321.562143
min	1.000000	1.000000
25%	120.250000	1.000000
50%	260.500000	2.000000
75%	615.250000	5.000000
max	4949.000000	6327.000000

	filesPerCommit	commitCount
0	1	6327
1	2	2609
2	3	1214
3	4	799
4	5	552
5	6	401
6	7	323
7	8	264
8	9	195
9	10	180
10	11	149
11	12	130
12	13	148
13	14	166
14	15	207
15	16	96
16	17	88
17	18	85
18	19	104
19	20	84
20	21	59
21	22	84
22	23	50
23	24	53
24	25	62
25	26	46
26	27	48
27	28	37
28	29	42
29	30	34

Bar chart with the number of files per commit distribution

Changed files per commit



Pairwise Changed Files

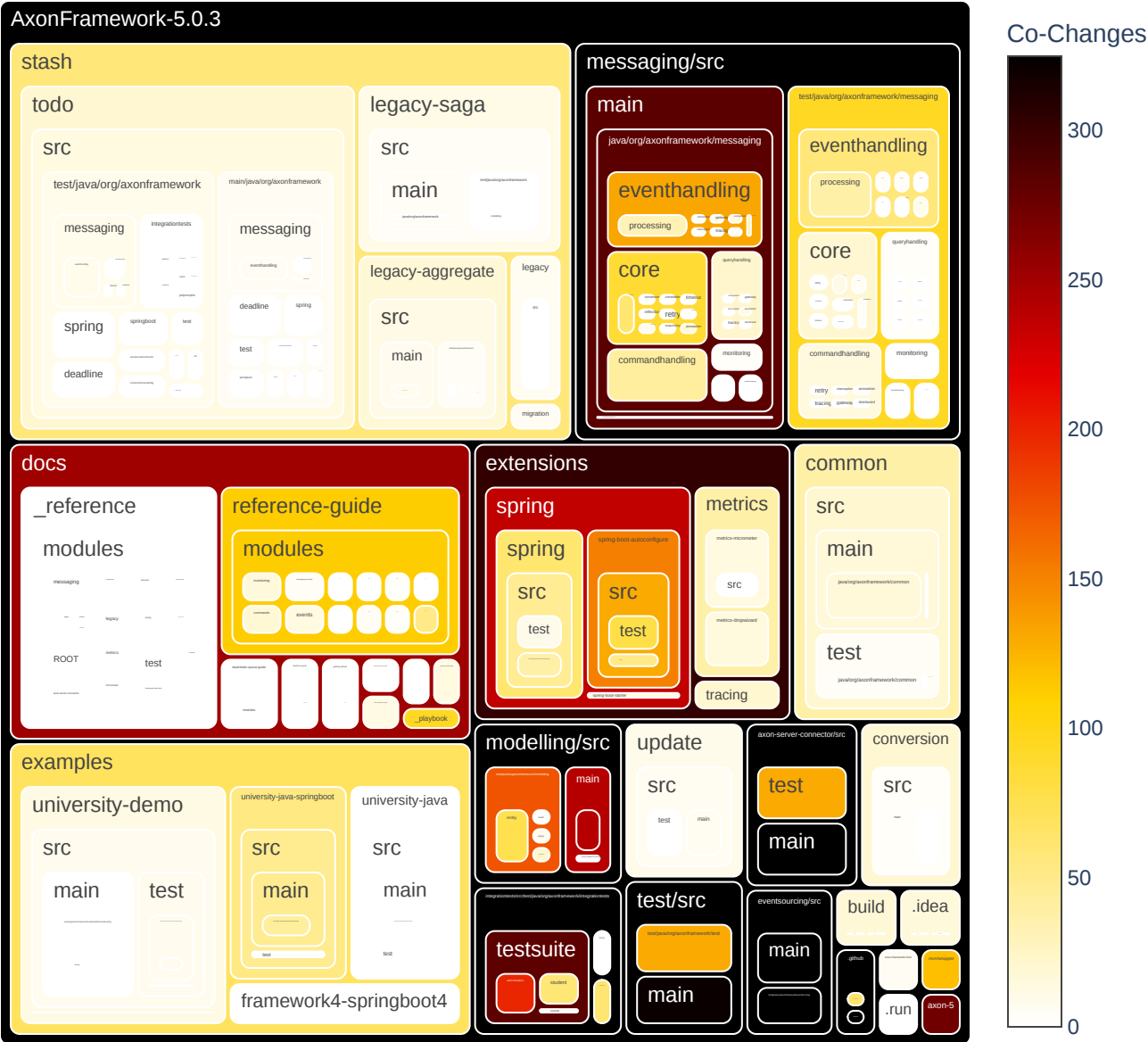
This section analyzes files that were changed together within the same commit and provides several metrics to quantify the strength of the co-change relationship:

- **Commit Count:** The number of commits in which two files were changed together.
- **Commit Lift:** A ratio that indicates whether the co-change pattern is stronger than random chance, given how often each file changes.
- **Jaccard Similarity:** The ratio of commits involving either file that also involved both files.

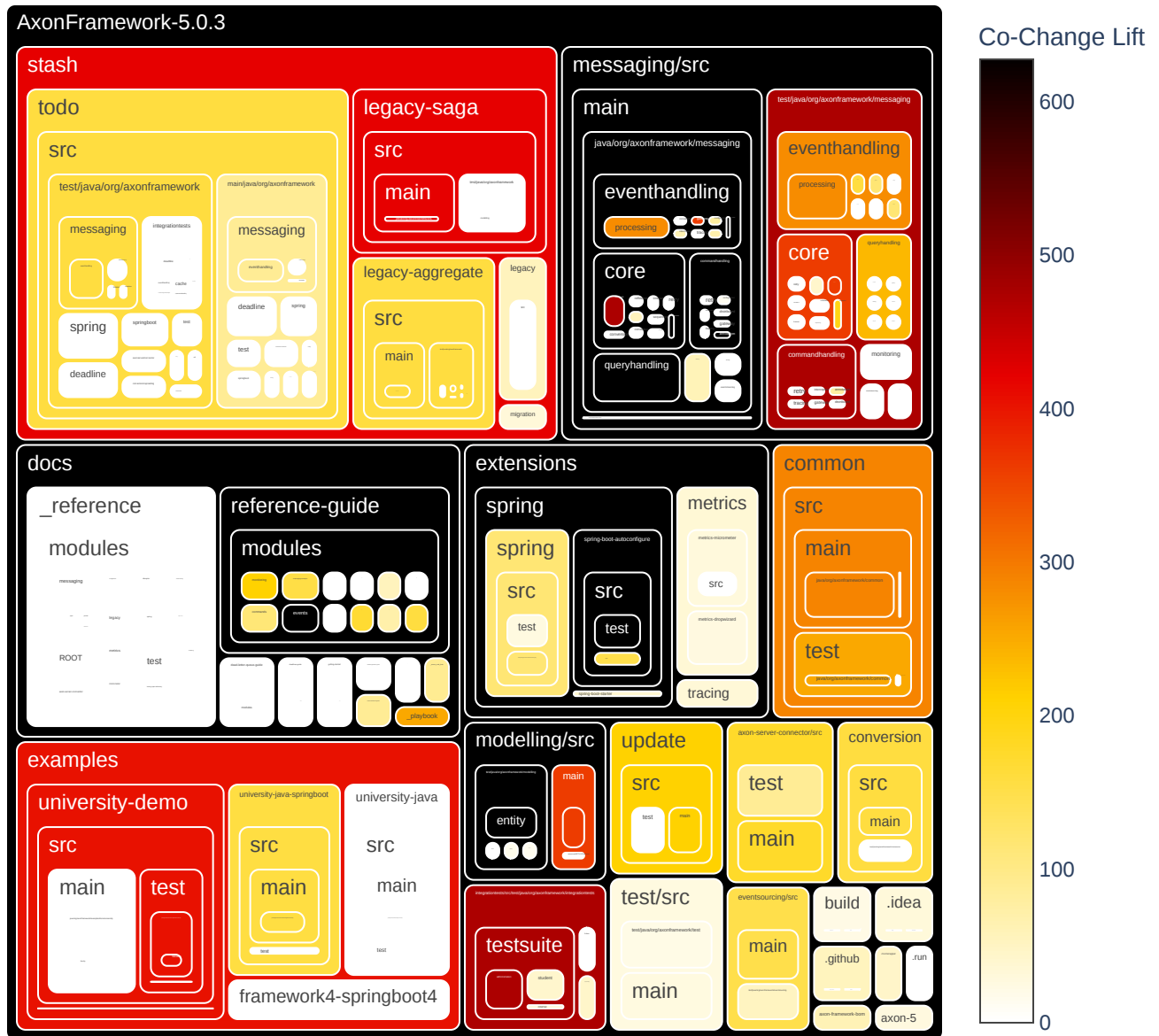
The following tables show the top pairwise changed files based on these metrics. The following charts show how these metrics are distributed across pairs of files that were changed together.

Treemap with files changed frequently with others

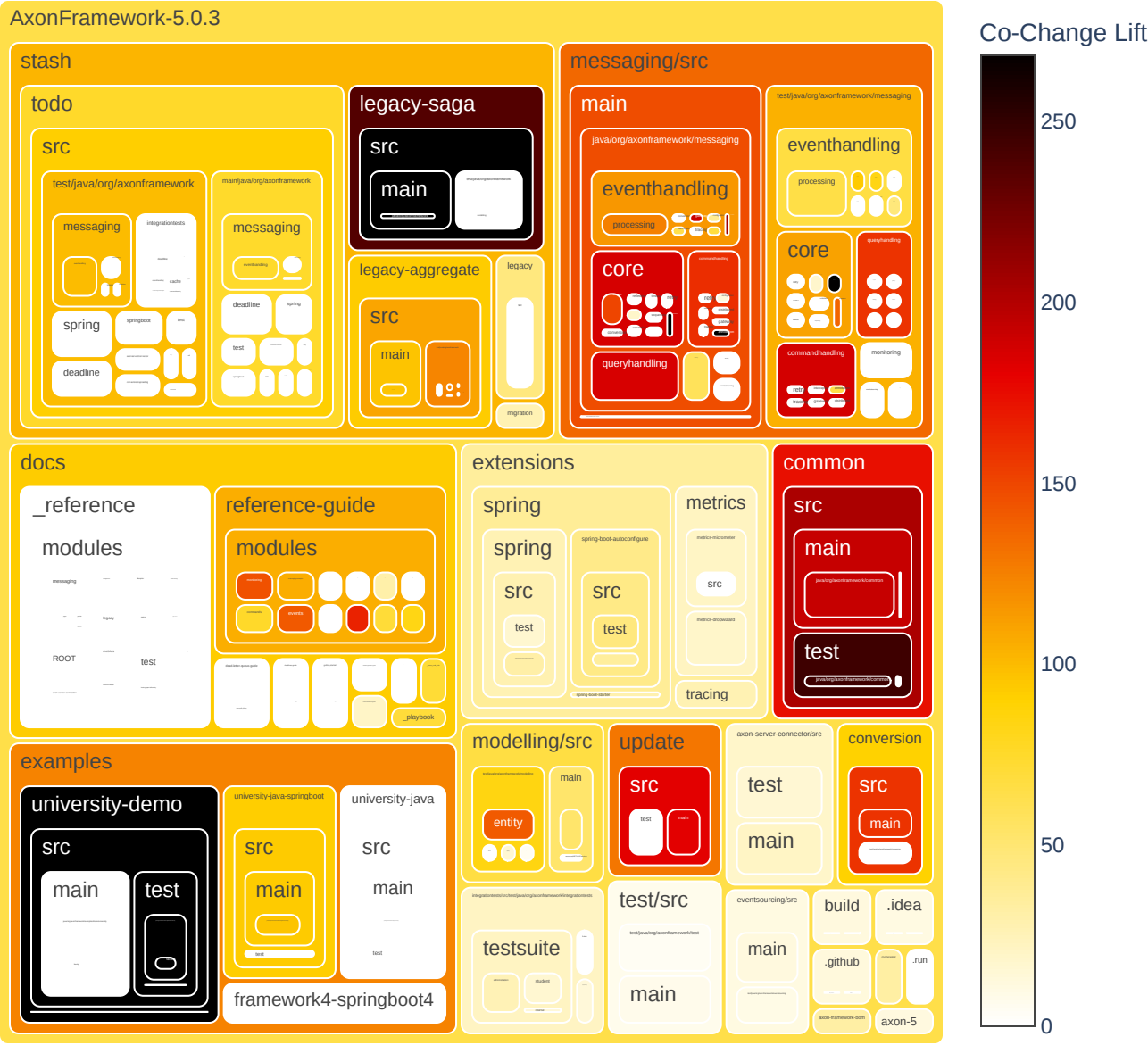
Files that likely co-change with others in update commits



Co-Changing files in update commits max lift (1=random, >1=more than random, <1=



Co-Changing files in update commits average lift (1=random, >1=more than random,

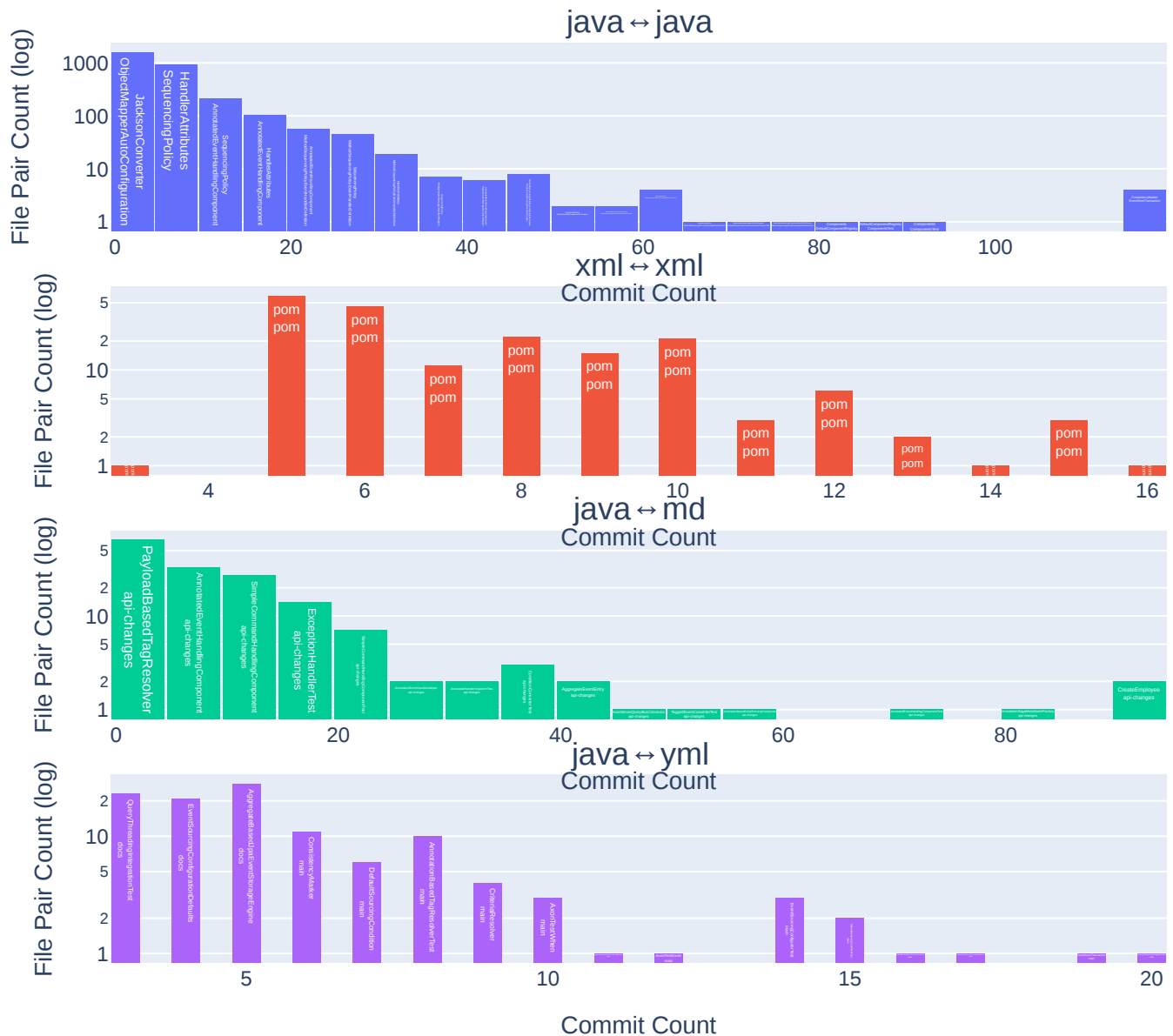


	fileExtensionPair	fileExtensionPairCount
0	java↔java	3005
1	xml↔xml	191
2	java↔md	162
3	java↔yml	117

Files changed together by commit count

	fileExtensionPair	updateCommitCount	GroupRank	filePair
0	java↔java	119	1	AggregateBasedStorageEngineTestSuite↔StorageEngineTestSuite
1	java↔java	117	2	AggregateBasedAxonServerEventStorageEngine↔AggregateBasedJpaEventStorageEngine connector
2	java↔java	116	3	AggregateBasedStorageEngineTestSuite↔AggregateBasedJpaEventStorageEngine event
3	java↔java	115	4	AggregateBasedStorageEngineTestSuite↔AggregateBasedAxonServerEventStorageEngine
4	java↔java	90	5	StorageEngineTestSuite↔AggregateBasedJpaEventStorageEngine
5	java↔java	88	6	StorageEngineTestSuite↔AggregateBasedAxonServerEventStorageEngine event
6	java↔java	80	7	StorageEngineTestSuite↔DefaultEventStoreTransactionTest
7	java↔java	77	8	AggregateBasedStorageEngineTestSuite↔DefaultEventStoreTransactionTest
8	java↔java	74	9	DefaultEventStoreTransaction↔StorageEngineTestSuite
9	java↔java	69	10	DefaultEventStoreTransaction↔AggregateBasedStorageEngineTestSuite
10	xml↔xml	16	1	pom↔pom
11	xml↔xml	15	2	pom↔pom
12	xml↔xml	14	3	pom↔pom
13	xml↔xml	13	4	pom↔pom
14	xml↔xml	12	5	pom↔pom
15	xml↔xml	11	6	pom↔pom
16	xml↔xml	10	7	pom↔pom
17	xml↔xml	9	8	pom↔pom
18	xml↔xml	8	9	pom↔pom
19	xml↔xml	7	10	pom↔pom
20	java↔md	92	1	AggregateBasedStorageEngineTestSuite↔api-changes
21	java↔md	91	2	AggregateBasedAxonServerEventStorageEngine↔api-changes
22	java↔md	82	3	AggregateBasedJpaEventStorageEngine↔api-changes
23	java↔md	74	4	StorageEngineTestSuite↔api-changes
24	java↔md	58	5	SimpleEventStoreTest↔api-changes
25	java↔md	54	6	DefaultEventStoreTransactionTest↔api-changes
26	java↔md	48	7	QueryThreadingIntegrationTest↔api-changes
27	java↔md	42	8	EventSourcingConfigurationDefaults↔api-changes
28	java↔md	38	9	AxonTestFixtureStatefulCommandHandlerTest↔api-changes
29	java↔md	36	10	AxonServerEventStorageEngine↔api-changes
30	java↔yaml	20	1	DefaultEventStoreTransaction↔pullrequest
31	java↔yaml	19	2	StorageEngineTestSuite↔pullrequest
32	java↔yaml	17	3	AggregateBasedStorageEngineTestSuite↔pullrequest
33	java↔yaml	16	4	AggregateBasedStorageEngineTestSuite↔main
34	java↔yaml	15	5	StorageEngineTestSuite↔main
35	java↔yaml	14	6	QueryThreadingIntegrationTest↔main
36	java↔yaml	12	7	NoAppendConditionTest↔pullrequest
37	java↔yaml	11	8	AppendCondition↔pullrequest
38	java↔yaml	10	9	NoAppendCondition↔main
39	java↔yaml	9	10	SimpleEventStoreTest↔main

Co-Changed Files by their commit count

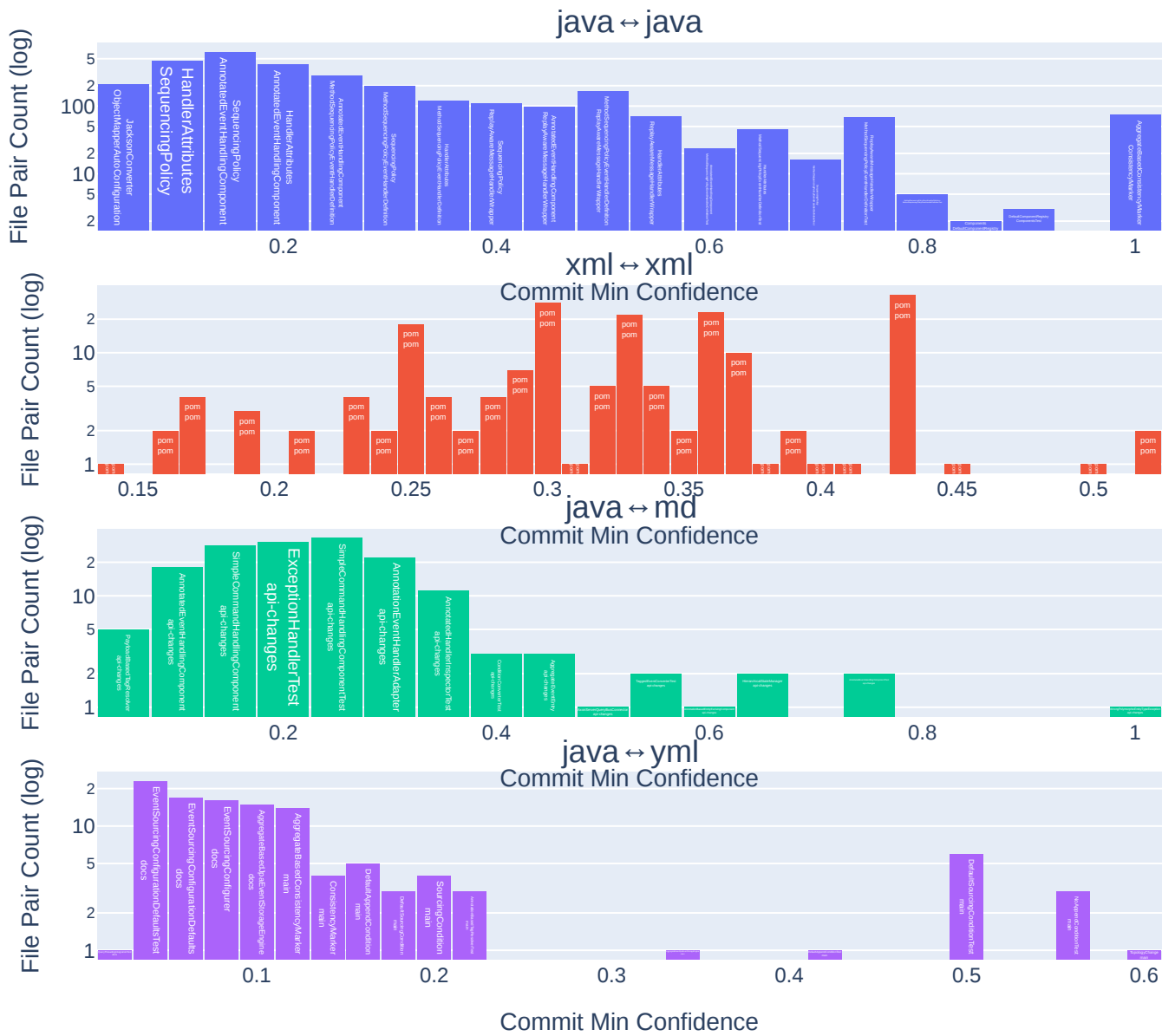


Files changed together by commit min confidence

The commit min confidence is the commit count where both files were changed divided by the commit count of the file with the least commits. This metric is useful to identify pairs of files that are frequently changed together and is not biased by single files that are changed very often.

	fileExtensionPair	updateCommitMinConfidence	GroupRank	filePair	
0	java↔java	1.000000	1	JacksonConverter↔ObjectMapperAutoConfiguration	con'
1	java↔java	0.900000	2	ChildEntityFieldDefinition↔GetterSetterChildEntityFieldDefinition	
2	java↔java	0.875000	3	ChildEntityFieldDefinition↔FieldChildEntityFieldDefinition	
3	java↔java	0.833333	4	AnnotatedEventHandlingComponent↔AnnotationEventHandlerAdapter	messaging/s
4	java↔java	0.800000	5	Components↔DefaultComponentRegistry	
5	java↔java	0.750000	6	FacultyAxonTestFixture↔CreateCourseAxonFixtureTest	
6	java↔java	0.727273	7	SimpleCommandBus↔TransactionalUnitOfWorkFactory	
7	java↔java	0.714286	8	DefaultAxonApplication↔FutureUtils	
8	java↔java	0.700000	9	NoTransactionManager↔TransactionManager	
9	java↔java	0.675000	10	AggregateBasedConsistencyMarker↔AggregateBasedStorageEngineTestSuite	e'
10	xml↔xml	0.517241	1	pom↔pom	
11	xml↔xml	0.500000	2	pom↔pom	
12	xml↔xml	0.450000	3	pom↔pom	
13	xml↔xml	0.428571	4	pom↔pom	
14	xml↔xml	0.413793	5	pom↔pom	
15	xml↔xml	0.400000	6	pom↔pom	
16	xml↔xml	0.392857	7	pom↔pom	
17	xml↔xml	0.378378	8	pom↔pom	
18	xml↔xml	0.371429	9	pom↔pom	
19	xml↔xml	0.370370	10	pom↔pom	
20	java↔md	1.000000	1	AnnotationSagaMetaModelFactory↔api-changes	
21	java↔md	0.750000	2	ImmutableReflectionEntityModelAdministrationIT↔api-changes	
22	java↔md	0.666667	3	AnnotationEventHandlerAdapter↔api-changes	
23	java↔md	0.600000	4	CommandConverter↔modularity	
24	java↔md	0.571429	5	ExceptionHandlerTest↔api-changes	
25	java↔md	0.500000	6	SimpleCommandHandlingComponentTest↔api-changes	
26	java↔md	0.471503	7	AggregateBasedAxonServerEventStorageEngine↔api-changes	
27	java↔md	0.470588	8	ChildAmbiguityException↔api-changes	
28	java↔md	0.434783	9	Repository↔api-changes	
29	java↔md	0.386792	10	AggregateBasedJpaEventStorageEngine↔api-changes	
30	java↔yaml	0.600000	1	SpringUtils↔slack-release-notification	
31	java↔yaml	0.555556	2	CreateCourseController↔application	
32	java↔yaml	0.500000	3	UnsubscribeStudentToCourseController↔application	e
33	java↔yaml	0.428571	4	EventProcessingAutoConfiguration↔test-docker-compose	
34	java↔yaml	0.333333	5	UpdateCheckerProperties↔application	
35	java↔yaml	0.222222	6	NoAppendCondition↔pullrequest	
36	java↔yaml	0.214286	7	GrpcMessageSizeInterceptor↔main	
37	java↔yaml	0.203390	8	NoAppendConditionTest↔pullrequest	
38	java↔yaml	0.200000	9	TopologyChange↔main	
39	java↔yaml	0.193548	10	DefaultEventStoreTransaction↔add-to-project	

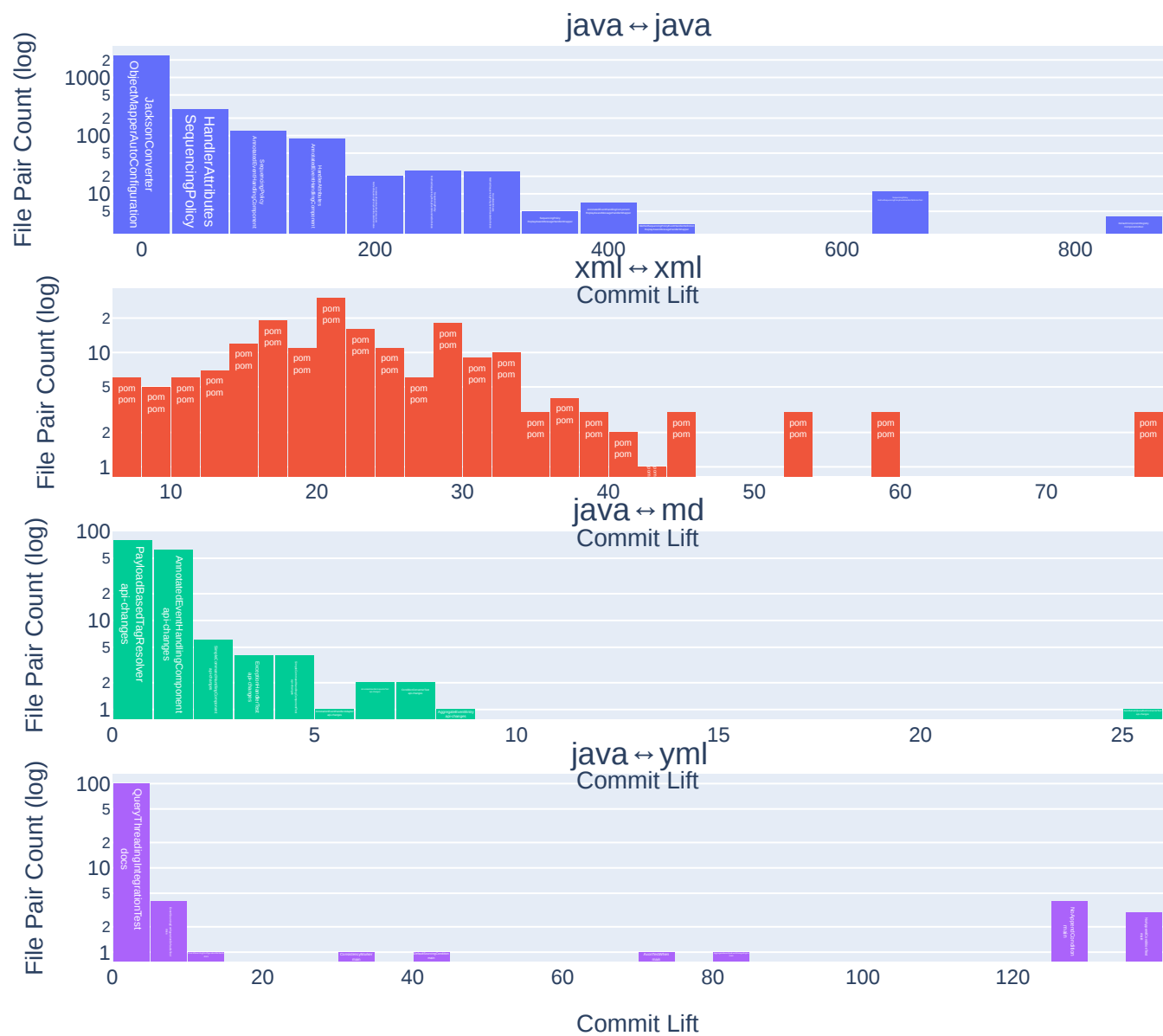
Co-Changed Files by their commit min confidence



Files changed together by commit lift

	fileExtensionPair	updateCommitLift	GroupRank	filePair
0	java↔java	836.333333	1	HandlerAttributes↔SequencingPolicy
1	java↔java	627.250000	2	Command↔AnnotationMessageTypeResolver
2	java↔java	470.437500	3	ImmutableReflectionEntityModelAdministrationIT↔MutableReflectionEntityModelAdministrationIT integr
3	java↔java	418.166667	4	HandlerComparator↔AnnotationSagaMetaModelFactory
4	java↔java	401.440000	5	CreateCourseAxonFixtureTest↔CreateCourseAxonFixtureTest
5	java↔java	376.350000	6	FacultyAxonTestFixtureTest↔CreateCourseAxonFixtureTest
6	java↔java	358.428571	7	SequencingPolicy↔ReplayAwareMessageHandlerWrapper
7	java↔java	313.625000	8	EventAppender↔EventPublishingUtils
8	java↔java	286.742857	9	DefaultComponentRegistry↔ComponentsTest
9	java↔java	278.777778	10	MergeTask↔SplitTask
10	xml↔xml	76.806122	1	pom↔pom
11	xml↔xml	59.738095	2	pom↔pom
12	xml↔xml	53.764286	3	pom↔pom
13	xml↔xml	44.803571	4	pom↔pom
14	xml↔xml	43.258621	5	pom↔pom
15	xml↔xml	41.816667	6	pom↔pom
16	xml↔xml	38.403061	7	pom↔pom
17	xml↔xml	37.078818	8	pom↔pom
18	xml↔xml	36.048851	9	pom↔pom
19	xml↔xml	35.842857	10	pom↔pom
20	java↔md	25.955172	1	CommandConverter↔modularity
21	java↔md	8.177966	2	NoAppendConditionTest↔design-principles
22	java↔md	7.658730	3	NoAppendCondition↔design-principles
23	java↔md	7.148148	4	DefaultAppendConditionTest↔design-principles
24	java↔md	6.433333	5	AppendCondition↔design-principles
25	java↔md	6.174273	6	QueryThreadingIntegrationTest↔README
26	java↔md	5.462264	7	AxonServerMessageStream↔design-principles
27	java↔md	4.453846	8	DefaultSourcingConditionTest↔design-principles
28	java↔md	4.386364	9	SourcingConditionTest↔design-principles
29	java↔md	4.147107	10	AnnotationSagaMetaModelFactory↔api-changes
30	java↔yaml	139.388889	1	CreateCourseController↔application
31	java↔yaml	125.450000	2	UnsubscribeStudentToCourseController↔application
32	java↔yaml	83.633333	3	UpdateCheckerProperties↔application
33	java↔yaml	73.794118	4	UpdateCheckerAutoConfiguration↔application
34	java↔yaml	40.467742	5	EventProcessingAutoConfiguration↔application
35	java↔yaml	34.686636	6	EventProcessingAutoConfiguration↔test-docker-compose
36	java↔yaml	12.866667	7	SpringUtils↔slack-release-notification
37	java↔yaml	6.858939	8	NoAppendConditionTest↔add-to-project
38	java↔yaml	6.744624	9	AppendCondition↔add-to-project
39	java↔yaml	6.423451	10	NoAppendCondition↔add-to-project

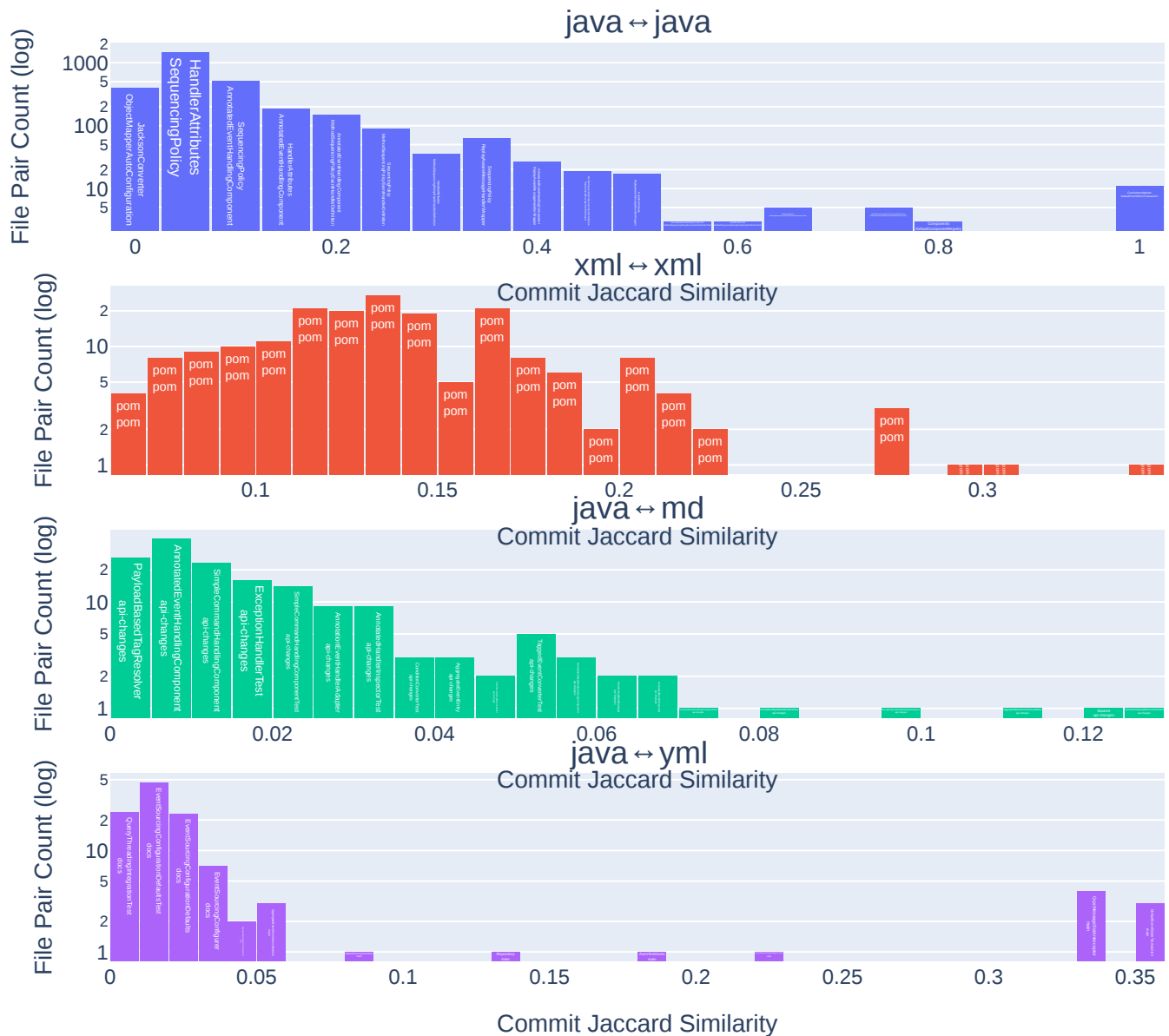
Co-Changed Files by their commit lift



Files changed together by commit Jaccard similarity

	fileExtensionPair	updateCommitJaccardSimilarity	GroupRank	1
0	java↔java	1.000000	1	HandlerAttributes↔Sequencir
1	java↔java	0.818182	2	ChildEntityFieldDefinition↔GetterSetterChildEntityFieldDe
2	java↔java	0.800000	3	FieldChildEntityFieldDefinition↔GetterSetterChildEntityFieldDe
3	java↔java	0.750000	4	Command↔AnnotationMessageTypeR
4	java↔java	0.666667	5	CreateCourseAxonFixtureTest↔CreateCourseAxonFixt
5	java↔java	0.636364	6	ChildEntityFieldDefinition↔FieldChildEntityFieldDe
6	java↔java	0.600000	7	ImmutableReflectionEntityModelAdministrationIT↔MutableReflectionEntityModelAdminist
7	java↔java	0.571429	8	AnnotationBasedEntityEvolvingComponent↔AnnotationSagaMetaModel
8	java↔java	0.555556	9	SimpleUnitOfWorkFactory↔UnitOfWorkConfig
9	java↔java	0.500000	10	DefaultComponentRegistry↔Compone
10	xml↔xml	0.340909	1	por
11	xml↔xml	0.300000	2	por
12	xml↔xml	0.294118	3	por
13	xml↔xml	0.272727	4	por
14	xml↔xml	0.227273	5	por
15	xml↔xml	0.222222	6	por
16	xml↔xml	0.215385	7	por
17	xml↔xml	0.214286	8	por
18	xml↔xml	0.208333	9	por
19	xml↔xml	0.204082	10	por
20	java↔md	0.128713	1	AggregateBasedAxonServerEventStorageEngine↔api-c
21	java↔md	0.120419	2	AggregateBasedStorageEngineTestSuite↔api-c
22	java↔md	0.111565	3	AggregateBasedJpaEventStorageEngine↔api-c
23	java↔md	0.095484	4	StorageEngineTestSuite↔api-c
24	java↔md	0.080556	5	SimpleEventStoreTest↔api-c
25	java↔md	0.074896	6	DefaultEventStoreTransactionTest↔api-c
26	java↔md	0.067989	7	QueryThreadingIntegrationTest↔api-c
27	java↔md	0.065476	8	QueryThreadingIntegrationTest↔R
28	java↔md	0.062500	9	NoAppendConditionTest↔design-pr
29	java↔md	0.061224	10	QueryThreadingIntegrationTest↔ISSUE_TEI
30	java↔yaml	0.357143	1	CreateCourseController↔app
31	java↔yaml	0.333333	2	UnsubscribeStudentToCourseController↔app
32	java↔yaml	0.227273	3	UpdateCheckerAutoConfiguration↔app
33	java↔yaml	0.187500	4	UpdateCheckerProperties↔app
34	java↔yaml	0.138889	5	EventProcessingAutoConfiguration↔app
35	java↔yaml	0.085714	6	EventProcessingAutoConfiguration↔test-docker-ci
36	java↔yaml	0.058824	7	NoAppendConditionTest↔add-to
37	java↔yaml	0.058140	8	AppendCondition↔add-to
38	java↔yaml	0.056180	9	NoAppendCondition↔add-to
39	java↔yaml	0.049383	10	DefaultAppendConditionTest↔add-to

Co-Changed Files by their commit jaccard similarity



Find pairwise changed files with many highly ranked metrics

Find those pairwise changed files that have a high rank in many metrics by calculating a combined (weighted) score based on the ranks of each metric. This is useful to identify pairs of files that score high in most metrics, which indicates a strong co-change relationship.

	fileExtensionPair	filePair	combinedMetricsScore	updateCommitCountEx
0	java↔java	HandlerAttributes↔SequencingPolicy	62	
1	java↔java	Command↔Event	62	
2	java↔java	Command↔Query	62	
3	java↔java	Event↔Query	62	
4	java↔java	Query↔QueryResponse	62	
5	java↔java	Command↔QueryResponse	62	
6	java↔java	Event↔QueryResponse	62	
7	java↔java	FieldChildEntityFieldDefinitionTest↔GetterEvolverChildEntityFieldDefinitionTest	62	
8	java↔java	GetterEvolverChildEntityFieldDefinitionTest↔GetterSetterChildEntityFieldDefinitionTest	62	
9	java↔java	FieldChildEntityFieldDefinitionTest↔GetterSetterChildEntityFieldDefinitionTest	62	
10	java↔md	AggregateBasedAxonServerEventStorageEngine↔api-changes	29	
11	java↔md	AggregateBasedJpaEventStorageEngine↔api-changes	38	
12	java↔md	AggregateBasedStorageEngineTestSuite↔api-changes	42	
13	java↔md	QueryThreadingIntegrationTest↔README	55	
14	java↔md	SimpleEventStoreTest↔api-changes	59	
15	java↔md	CommandConverter↔modularity	59	
16	java↔md	AxonServerEventStorageEngine↔api-changes	67	
17	java↔md	QueryThreadingIntegrationTest↔api-changes	70	
18	java↔md	DefaultEventStoreTransactionTest↔api-changes	72	
19	java↔md	StorageEngineTestSuite↔api-changes	74	
20	java↔yaml	CreateCourseController↔application	18	
21	java↔yaml	RenameCourseController↔application	18	
22	java↔yaml	ChangeCourseCapacityController↔application	18	
23	java↔yaml	UnsubscribeStudentToCourseController↔application	21	
24	java↔yaml	EnrollStudentInFacultyController↔application	21	
25	java↔yaml	CourseStatsController↔application	21	
26	java↔yaml	SubscribeStudentToCourseController↔application	21	
27	java↔yaml	UpdateCheckerAutoConfiguration↔application	24	
28	java↔yaml	EventProcessingAutoConfiguration↔application	27	
29	java↔yaml	UpdateCheckerProperties↔application	28	
30	xml↔xml	pom↔pom	9	
31	xml↔xml	pom↔pom	14	
32	xml↔xml	pom↔pom	18	
33	xml↔xml	pom↔pom	20	
34	xml↔xml	pom↔pom	20	
35	xml↔xml	pom↔pom	20	
36	xml↔xml	pom↔pom	26	
37	xml↔xml	pom↔pom	26	
38	xml↔xml	pom↔pom	26	
39	xml↔xml	pom↔pom	34	

Pairwise changed files with pareto-optimal metrics

A pair (count, confidence, jaccard, lift) is Pareto-optimal if there is no other pair that is better or equal in all metrics and strictly better in at least one. In other words, it is not "dominated" by any other pair.

The frontier = the “best tradeoffs.”

Pairwise changed files with pareto-optimal metrics - not considering file extensions

	filePair	combinedMetricsScore	updateCommitCount	updateCommitH
0	HandlerAttributes↔SequencingPolicy	62	3	
1	AggregateBasedConsistencyMarker↔AggregateBasedStorageEngineTestSuite	1106	27	
2	AggregateBasedStorageEngineTestSuite↔StorageEngineTestSuite	1062	119	
3	pom↔pom	9	15	
4	AxonTestGiven↔AxonTestWhen	442	48	
5	ConsistencyMarker↔AbstractConsistencyMarker	413	10	
6	ConsistencyMarker↔ConsistencyMarkers	341	17	
7	AggregateBasedAxonServerEventStorageEngine↔AggregateBasedJpaEventStorageEngine	789	117	
8	EventSourcingConfigurationDefaults↔EventSourcingConfigurationDefaultsTest	861	55	
9	DefaultAppendConditionTest↔NoAppendConditionTest	269	35	
10	DefaultSourcingConditionTest↔SourcingConditionTest	279	43	
11	Command↔Event	62	4	
12	AnnotatedCommandHandlingComponent↔AnnotatedQueryHandlingComponent	241	10	
13	Command↔Query	62	4	
14	Event↔Query	62	4	
15	Query↔QueryResponse	62	4	
16	Command↔QueryResponse	62	4	
17	Event↔QueryResponse	62	4	
18	FieldChildEntityFieldDefinition↔GetterEvolverChildEntityFieldDefinition	64	8	
19	ChildEntityFieldDefinition↔GetterSetterChildEntityFieldDefinition	75	9	
20	FieldChildEntityFieldDefinitionTest↔GetterEvolverChildEntityFieldDefinitionTest	62	3	
21	GetterEvolverChildEntityFieldDefinitionTest↔GetterSetterChildEntityFieldDefinitionTest	62	3	
22	FieldChildEntityFieldDefinitionTest↔GetterSetterChildEntityFieldDefinitionTest	62	3	

Pairwise changed files with pareto-optimal metrics - using ranks grouped by file extensions

	fileExtensionPair	filePair	combinedMetricsScore	updateCommitCount	updateCommitH
0	java↔java	DefaultEventStoreTransaction↔AxonTestGiven	3587	3	
1	java↔java	QueryThreadingIntegrationTest↔EventSourcingConfigurationDefaults	3580	4	
2	java↔java	AggregateBasedJpaEventStorageEngine↔EventSourcingConfigurationDefaultsTest	3589	4	
3	java↔java	AggregateBasedAxonServerEventStorageEngine↔ModellingConfigurer	3589	3	
4	java↔java	AggregateBasedJpaEventStorageEngine↔ModellingConfigurerTest	3600	3	
5	java↔java	AggregateBasedStorageEngineTestSuite↔RecordingEventSink	3608	3	

WordCloud of git authors

	word	frequency
0	Steven van Beelen	6483
1	Allard Buijze	2930
2	smcvb	822
3	Mitchell Herrijgers	741
4	Mateusz Nowak	733
5	Milan Savic	449
6	Gerard Klijs	246
7	Marc Gathier	231
8	Simon Zambrovski	227
9	Jan Galinski	174

